

Krishnaa Vadivel | Electrical Engineer | Semiconductor Devices | Photonics | FPGA Systems

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Profile

Electrical engineering PhD researcher with broad preparation across semiconductor physics, devices, integrated photonics, embedded systems, digital design, and scientific software. Combines theory-level understanding with implementation experience in FPGA/MCU systems, numerical device modeling, accelerator-enabled computation, and engineering tooling.

Research Experience

Yale University, Electrical Engineering

Research Assistant / PhD Researcher

2021–present

- Conduct research connecting materials-scale and device-relevant modeling, including first-principles excited-state methods, semiconductor physics, many-body theory, and scalable computational workflows.
- Develop numerical and software infrastructure in Python and C++ for simulation and analysis on accelerator-enabled HPC systems, using MPI, OpenMP, CUDA, ROCm, PETSc, and SLEPc.
- Serve as an embedded systems teaching assistant, supporting laboratory instruction and student work involving firmware, hardware interfaces, debugging, and instrumentation.
- Maintain strong continuity between theory and engineering practice through coursework and self-developed notes in electromagnetism, signals/systems, semiconductor transport, optics, and device equations.
- Co-authored a 2025 *Journal of the American Chemical Society* paper and presented related APS work in 2024, 2025, and 2026 on self-trapped excitons and excited-state materials phenomena.

Professional Experience

Probe Technologies Inc. / Weatherford Wireline Services

Software and Embedded Systems Engineer

2019–2021

- Developed firmware and digital logic for sensor-logging and control functions in geological well tools using Lattice FPGA, PIC32, dsPIC, STM32, and C8051-class hardware.
- Built CUDA-based processing and custom visualization tools for large acoustic waveform datasets used in engineering analysis and field diagnostics.
- Contributed host-side software and infrastructure using C#, ASP-based services, and Microsoft SQL-backed data workflows.
- Participated in board- and system-level engineering tasks including selective Altium PCB updates, oscilloscope-based verification, integration testing, and field deployment support.

FindLight

Photonics Technical Writer

2018–2019

- Produced technical articles on lasers, photonics components, optical systems, and engineering applications for a specialized industrial audience.

Selected Projects

FPGA AI Accelerator for ML Datapaths

- Implemented a compact Verilog matrix-multiply accelerator and verification flow to demonstrate datapath tradeoffs relevant to AI hardware acceleration.

Single-Cycle MIPS Processor

- Built a small Verilog CPU and self-checking Icarus Verilog test bench to demonstrate ISA-level control, datapath design, and reproducible digital verification.

Semiconductor Device Simulation

- Developed numerical treatments of Poisson, drift-diffusion, and pn-junction-style device equations to connect semiconductor theory with engineering-oriented simulation.

Integrated Photonics and FDTD Modeling

- Modeled resonator and band-pass style photonic structures with Meep-based FDTD workflows for integrated-optics exploration.

Robotic Hand and Embedded Robotics Work

- Built a surface-EMG-controlled robotic hand in undergraduate Cornell ECE work, integrating analog filtering, PIC32 classification logic, and servo actuation; related work was published in Circuit Cellar.

Maze Mapping Robot

- Worked on FPGA- and algorithm-driven robotic navigation with wireless control and remote-station coordination in course-based robotics work.

Education

Yale University

PhD, Electrical Engineering

2021–present

Ongoing research spanning computational materials, semiconductor-relevant modeling, and scientific software.

Yale University

MPhil, Electrical Engineering

2023

Yale University

MS, Electrical Engineering

2023

Cornell University

MEng, Electrical and Computer Engineering

2017–2018

Included work in stochastic computing, semiconductor/device topics, and advanced EE design.

Cornell University

BS, Electrical and Computer Engineering

2013–2017

Included strong preparation in embedded systems, robotics, controls, and hardware-software integration.

Selected Coursework

Semiconductors: Semiconductor Physics; Thin Film Science; drift-diffusion and device modeling foundations

Photonics: Lasers and Optoelectronics; Integrated Photonics

Circuits and VLSI: VLSI Design; Analog VLSI Design; stochastic-computing-related design work

Systems: Mathematics of Signals and Systems; Advanced Embedded Systems

Foundations: Graduate Quantum Mechanics; Solid State Physics; Many-Body Quantum Physics

Technical Skills

Languages: Python, C, C++, CUDA, JavaScript, Bash, PowerShell

Hardware: Verilog, FPGA, PIC32, dsPIC, STM32, C8051, sensor interfaces, PCB bring-up, oscilloscope debugging

Simulation: Semiconductor device modeling, FDTD photonics simulation, numerical PDE workflows

Software: C#, ASP.NET, Microsoft SQL Server, Linux command line

Scientific Computing: MPI, OpenMP, PETSc, SLEPc, NumPy, pandas, SciPy, SymPy, matplotlib

Platforms: Docker, Docker Compose, Kubernetes, gcloud, Frontier, Frontera, Perlmutter